

Technology #10

Autonomous Catheter Navigation with
Human-in-the-Loop

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Autonomous Catheter Navigation with Human-in-the-Loop

WHY DOES IT MATTER?

Catheter placement is operator-dependent; malposition causes complications

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 5 governing equations, 11 design inputs, 7 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Catheter Mechanics + Hydraulic Actuation + Tissue Interaction. 6-step build plan with specific test articles and acceptance criteria. Target buyer: Surgical robotics company; neuroshunt manufacturer.

WHAT DOES THE BUYER GET?

Autonomous navigation concept + R1 correction (buckling + tissue safety + human-in-the-loop); Beam theory + buckling + hydraulic governing equations; Bench test protocols (buckling, navigation accuracy, tissue damage, biocompatibility); External precedent catalog (steerable catheter literature); Critical UNKNOWN disclosure (tissue damage threshold, achievable accuracy); This engineering dossier with failure analysis + build plan

WHAT DOES THE BUYER HAVE TO BUILD?

Production catheter design (multi-lumen, actuator integrated); Production navigation controller (software V&V; per IEC 62304); Manufacturing process for steerable shunt catheter; Regulatory submission (autonomous navigation adds complexity; likely Class III); Clinical validation (placement accuracy endpoint); Tissue damage threshold establishment (animal studies)

WHAT IS THE NEXT DECISIVE EXPERIMENT?

Catheter mechanical specimens (various EI)

WHAT WOULD MAKE US KILL IT?

Buckling threshold is below typical insertion forces (catheter cannot navigate without buckling)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 6-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: Catheter mechanical specimens (various EI)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - Catheter mechanical specimens (various EI) - 10 weeks

Basis: Build plan WP. Scope: EI, buckling threshold P_{cr} . Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - Anatomical phantom - 12 weeks

Basis: Build plan WP. Scope: Navigation accuracy + contact forces. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Biocompatibility specimens (Pebax multi-lumen catheter + hydraulic actuator silicone) - 12 weeks (external)

Basis: Build plan WP. Scope: ISO 10993 series. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 5 equations. Key assumptions: Catheter is slender beam ($L \gg D$); Material is linear elastic (small strain); Actuator response quasi-static; Tissue is elastic with damage threshold (UNKNOWN specific value); Anatomy known from preoperative imaging. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: Catheter mechanical specimens (various EI). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Hydraulic steering with closed-loop navigation; human-in-the-loop fallback (R1 fix); buckling analysis

POTENTIAL VALUE DRIVER: Addresses: Catheter placement is operator-dependent; malposition causes complications

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Surgical robotics company; neuroshunt manufacturer

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Catheter Mechanics + Hydraulic Actuation + Tissue Interaction

Disciplines: Mechanical engineering (catheter mechanics, buckling), Hydraulic actuation, Tissue biomechanics, Control systems (closed-loop navigation), Sterilization and biocompatibility

Architecture: Steerable CSF shunt catheter with hydraulic actuation for autonomous navigation during placement. R1 fix: original P-22 assumed fully autonomous navigation without addressing buckling failure modes; R1 redesign adds buckling analysis, tissue damage safety, and human-in-the-loop fallback.

Subsystems:

SS-01: Catheter body (multi-lumen, flexible) - Carries CSF flow + houses steering lumens + sensor channels

SS-02: Hydraulic steering actuator - Inflatable/deflatable balloon or bellows at catheter tip; bends catheter when pressurized

SS-03: Tip position sensor - Measures catheter tip position and orientation for closed-loop navigation

SS-04: Navigation controller - Closed-loop controller computing actuator pressures to track target trajectory; human-in-the-loop fallback (R1 fix)

2. Mechanism Architecture

Physical Changes: Hydraulic pressure in steering lumen deflects catheter tip; coordinated multi-lumen pressure changes navigate catheter through anatomy

Key Physics: Catheter bending governed by beam theory: curvature $\kappa = M / (E \cdot I)$, where M is bending moment from actuator force. Buckling failure when axial compression exceeds Euler load $P_{cr} = \pi^2 \cdot E \cdot I / L^2$.

3. Governing Engineering Model

Summary: Hydraulic beam steering with buckling constraint. Catheter is modeled as flexible beam with hydraulically-actuated bending moment; axial compression limited by Euler buckling.

Equations:

$P_{cr} = (\pi^2 \cdot E \cdot I) / L^2$ [elastic column buckling]

$\kappa = M / (E \cdot I)$ [beam curvature]

$F_{hydraulic} = P_{actuator} \cdot A_{actuator}$ [actuator force]

$M_{tip} = F_{hydraulic} \cdot d_{moment_arm}$ [bending moment at tip]

$F_{contact_tissue} < F_{damage_threshold}$ [tissue safety constraint]

Assumptions:

- Catheter is slender beam ($L \gg D$)
- Material is linear elastic (small strain)
- Actuator response quasi-static
- Tissue is elastic with damage threshold (UNKNOWN specific value)
- Anatomy known from preoperative imaging

Boundary Conditions:

- Proximal: insertion force applied by clinician/robot
- Distal: free tip with actuator moment
- Wall contact: tissue reaction force
- Axial compression $< P_{cr}$ (buckling constraint)

Failure Regimes:

- Buckling (axial force $> P_{cr}$) — catheter collapses

- Tissue damage (contact force $>$ threshold) — puncture or trauma
- Actuator saturation — cannot achieve required bending
- Navigation error — trajectory diverges from target
- Tissue property variation — model mismatch

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Catheter placement accuracy affects shunt function; current placement is manual	VERIFIED Operator-dependent (R332)
DI-002	Functional requirement	Autonomous catheter navigation via hydraulic actuation; R1 fix: bounded by model	MODELLED Issue safety + human-in-the-loop
DI-003	Anatomical trajectory	Cranial to peritoneal; ~30-50 cm typical	VERIFIED
DI-004	Catheter diameter	Constrained to ~2 mm (standard shunt catheter)	VERIFIED
DI-005	Buckling constraint	Axial force < $P_{cr} = \pi^2 EI / L^2$	MODELLED
DI-006	Tissue damage threshold	UNKNOWN — tissue-specific	UNKNOWN
DI-007	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-008	Sterilization	UNKNOWN	UNKNOWN
DI-009	Navigation accuracy target	UNKNOWN — clinical utility threshold TBD	UNKNOWN
DI-010	Human-in-the-loop requirement (R1 fix)	Operator must be able to override or abort navigation (R1 fix)	MODELLED
DI-011	EMC (if electronic controller)	UNKNOWN	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Multi-lumen catheter body design (geometry, materials, bends profile)	ABSENT	EI target; Material selection; Multi-lumen layout
DO-002	Hydraulic actuator design (balloon or bellows)	ABSENT	Actuator area; Pressure range; Material
DO-003	Tip position sensor	CONCEPTUAL	
DO-004	Navigation controller	MODELLED	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
Flexural rigidity EI	UNKNOWN — design choice based on bending stiffness	N·m ²	Material properties & geometry	Bench bending test per ISO 10555-1
Catheter length L	UNKNOWN — anatomy-dependent (cranial to pelvic)	cm	Anatomical data (typical 30-50 cm)	Anatomical measurement to confirm range
Actuator pressure P	UNKNOWN	Pa	Actuator design	Pressure test
Actuator area A	UNKNOWN	m ²	Geometry	
Buckling threshold P_cr	DERIVED from EI and L	N	Euler formula	Bench buckling test
Tissue damage threshold	UNKNOWN — varies by tissue type		UNKNOWN	Literature review + animal study
Tip position sensor accuracy	UNKNOWN	mm	Sensor selection	

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Buckling	Axial compression exceeds Euler load	Reduce P, reduce L; limit insertion force	Bench buckling test	UNKNOWN in vivo buckling threshold
Tissue damage	Contact force exceeds tissue damage threshold	Force sensing; soft tip; limit + alarm	Animal study with tissue damage	UNKNOWN tissue damage threshold
Navigation error	Controller diverges from target trajectory	Real-time loop fallback (Plan B)	Phantom trajectory test	UNKNOWN achievable accuracy
Actuator failure	Hydraulic leak or pump failure	Redundant actuator; manual fallback	Failure mode test	UNKNOWN failure rate
Sensor failure	Tip position sensor malfunction	Redundant sensing; imaging	Sensor fallback test	UNKNOWN failure rate
Anatomical variation	Patient anatomy differs from preoperative model	Intraoperative imaging update	Phantom with anatomical variation	UNKNOWN anatomical variation distribution
Catheter jamming	Mechanical obstruction in actuator	Flush protocol; redundant lumen	Jam simulation	UNKNOWN jam rate

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Buckling	Axial compression > Euler load	Increase EI; reduce L; limit insertion force	UNKNOWN monitoring threshold
Tissue damage	Contact force > damage threshold	Force sensing + limit + soft tip	CRITICAL UNKNOWN — threshold tissue-specific
Navigation error	Controller divergence	Human-in-the-loop fallback (R1 fix); abort	UNKNOWN achievable accuracy
Actuator failure (leak)	Hydraulic leak or pump failure	Redundant actuator; manual override	UNKNOWN failure rate
Sensor failure	Tip position sensor malfunction	Redundant sensing; imaging fallback	UNKNOWN failure rate
Anatomical variation	Patient anatomy differs from model	Intraoperative imaging update; human oversight	UNKNOWN variation distribution
Catheter jamming	Obstruction in actuator lumen	Flush protocol; redundant lumen	UNKNOWN jam rate

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Buckling threshold > expected forces	Bench buckling test	$P_{cr} \geq 2 \times \text{max expected force}$	NOT_TESTED
V-002	Contact force < damage threshold	Bench + tissue study	$F_{contact} < \text{threshold}$	NOT_TESTED
V-003	Navigation accuracy in phantom	Phantom navigation test	Targeting accuracy $\leq$ target	NOT_TESTED
V-004	Biocompatibility ISO 10993	ISO 10993	Pass	NOT_TESTED
V-005	ISO 10555-1 compliance	ISO 10555-1	Pass	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	Navigation accuracy in animal model	In vivo study (IACUC)	Clinically useful accuracy; no tissue damage	NOT_PERFORMED
VAL-002	Clinical placement accuracy	Clinical trial (IDE)	Non-inferior or superior to standard	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Catheter body	Pebax (polyether block amide)	Standard catheter material with ISO 10993	ISO 10993 + mechanical characterization	CANDIDATE
Catheter body	Polyurethane	Standard catheter material	ISO 10993 + mechanical	CANDIDATE
Hydraulic actuator	Silicone elastomer	Standard balloon material	ISO 10993 + fatigue + pressure	CANDIDATE
Reinforcement braid (if needed)	Stainless steel or nitinol braid	Standard catheter reinforcement	ISO 10993 + mechanical	CANDIDATE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	Multi-lumen catheter body	1	CUSTOM_COMPONENT	UNKNOWN — likely Pebax or Polyurethane (Standard catheter materials)	Multiple (NDI, etc)	MEDIUM	Material compatibility
02	Hydraulic actuator (balloon or bellows)	1 or less	CUSTOM_COMPONENT	UNKNOWN — likely silicone or polyurethane	Multiple (NDI, etc)	MEDIUM	Pressure + fatigue + leak test
03	Tip position sensor	1	COTS_CANDIDATE	NTA (COTS)	Multiple (NDI, etc)	HIGH	Calibration + integration
04	Hydraulic pump + valves	1 set	COTS_CANDIDATE	NTA (COTS)	Multiple	HIGH	Calibration + reliability
05	Controller (real-time MCU)	1	COTS_CANDIDATE	NTA (COTS)	Multiple (TI, STM)	HIGH	Software V&V per IEC 62304

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no autonomous steerable shunt catheter process validated

Candidate Processes:

- Multi-lumen extrusion (catheter body): Standard medical tubing process - Requires custom extrusion die
- Balloon/bellows fabrication (actuator): Standard balloon catheter manufacturing - Custom for small catheter size
- Catheter assembly (bonding, tipping): Standard catheter assembly - Requires assembly vendor

14. External Evidence

Source 1: A Soft-Tip Hydraulically Steerable Catheter for Enhanced

URL: <https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.202500099>

Excerpt: This study introduces a hydraulically steerable catheter with a soft tip in vascular procedures. The steering soft tip achieves a minimal ...

Source 2: Full article: Steerable catheters for minimally invasive surgery

URL: <https://www.tandfonline.com/doi/full/10.1080/24699322.2018.1526972>

Excerpt: by X Hu · 2018 · Cited by 217 — The catheters actuated by shape memory effects, conducting polymers and hydraulic pressure require relatively complex fabrication techniques ...

Source 3: Soft robotic steerable micro-catheter for the endovascular

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9809155>

Excerpt: by T Gopesh · 2021 · Cited by 172 — Controlled by hand, the devices offer complete 3D orientation of the tip. Using saline as a working fluid, we demonstrate guidewire-free navigation, access, and ...

Source 4: (PDF) A Soft-Tip Hydraulically Steerable Catheter for

URL: https://www.researchgate.net/publication/392919122_A_Soft-Tip_Hydraulically_Steerable_Catheter_for_Enhanced_Flexibility_and_Safety_in_Vascular_Interventions

Excerpt: Herein, a novel steerable catheter is presented, featuring a hydraulically actuated, soft, steerable tip that allows for real-time visualization ...

Source 5: Development of a smart guide wire using an electrostrictive

URL: <https://www.nature.com/articles/srep18593>

Excerpt: by F Ganet · 2015 · Cited by 62 — Shape memory polymers (SMPs) have polymer and exposing it to an alternating magnetic field, as could be used in conjunction with MRI12.

15. Transfer Boundary

YOU RECEIVE:

- + Autonomous navigation concept + R1 correction (buckling + tissue safety + human-in-the-loop)
- + Beam theory + buckling + hydraulic governing equations
- + Bench test protocols (buckling, navigation accuracy, tissue damage, biocompatibility)
- + External precedent catalog (steerable catheter literature)
- + Critical UNKNOWN disclosure (tissue damage threshold, achievable accuracy)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Production catheter design (multi-lumen, actuator integrated)
- > Production navigation controller (software V&V; per IEC 62304)
- > Manufacturing process for steerable shunt catheter
- > Regulatory submission (autonomous navigation adds complexity; likely Class III)
- > Clinical validation (placement accuracy endpoint)
- > Tissue damage threshold establishment (animal studies)

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Achievable EI balancing bendability and buckling resistance — UNKNOWN
- Tissue damage threshold for relevant anatomies — UNKNOWN (critical)
- Navigation accuracy in vivo — UNKNOWN
- Actuator failure rate — UNKNOWN
- Sensor failure rate — UNKNOWN
- Regulatory pathway (autonomous navigation adds complexity; likely Class III) — UNKNOWN
- Clinical placement accuracy requirement — UNKNOWN
- Anatomical variation distribution — UNKNOWN

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	Catheter mechanical properties	Specimen (vascular), buckling threshold	Force vs deflection	Force range; P _{crit}	Catheter mechanical properties	2 weeks
WP-02	Hydraulic actuator prototype	Pressure source, displacement transducer	Force vs deflection	Force range; P _{crit}	Hydraulic actuator prototype	4 weeks
WP-03	Anatomical phantom	Phantom, imaging, navigation system	Navigation accuracy	Accuracy vs target; Force vs target	Phantom navigation accuracy	2 weeks
WP-04	Tissue damage specimen (clinical)	Histology	Tissue damage vs force	Threshold established	Tissue damage threshold report	12 weeks
WP-05	Biocompatibility specimen (ISO 10993)	ISO 10993 series	Hydraulic actuator silicone	ISO 10993 report	ISO 10993 report	12 weeks (external)

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-06	In vivo navigation protocol	Animal facility, imaging	Navigation accuracy	Assembly clinically used	In vivo navigation	26 weeks

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 5 governing equations and a 6-step build plan. The next experiment is clearly specified. Target buyer: Surgical robotics company; neuroshunt manufacturer.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Buckling threshold is below typical insertion forces (catheter cannot navigate without buckling)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Buckling threshold is below typical insertion forces (catheter cannot navigate without buckling)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.